

Average Payoff, Typical Ticket

AP Statistics · Unit 2 · Topic 2.8 · TEACHER KEY

CED TETHER

- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU VAR-5** — Probability distributions may be used to model variation in populations.
- **LO VAR-5.A** — Represent the probability distribution for a discrete random variable. [Skill 2.B]
- **EK VAR-5.A.1** — The values of a random variable are the numerical outcomes of random behavior.
- **EK VAR-5.A.2** — A discrete random variable is a variable that can only take a countable number of values. Each value has a probability associated with it. The sum of the probabilities over all of the possible values must be 1.
- **EK VAR-5.A.3** — A probability distribution can be represented as a graph, table, or function showing the probabilities associated with values of a random variable.
- **EK VAR-5.A.4** — A cumulative probability distribution can be represented as a table or function showing the probability of being less than or equal to each value of the random variable.
- **LO VAR-5.B** — Interpret a probability distribution. [Skill 4.B]
- **EK VAR-5.B.1** — An interpretation of a probability distribution provides information about the shape, center, and spread of a population and allows one to make conclusions about the population of interest.

Essential question: How can an average payout and the most common outcome describe different parts of the same distribution?

DOK-3 skill: 4.B · **FRQ pattern:** expected-value-vs-typical-outcome-what-the-mean-tells-one-person

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Reconciles the given average payout with most tickets paying nothing

- **reasoning-2** (required) — Uses the 2-dollar cost and 87.5-percent loss chance to assess Priya’s claim
- **reasoning-3** (required) — Distinguishes an average from the possible result of one ticket

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Treating the average payout as the most common ticket outcome
- Ignoring the 2-dollar ticket cost
- Saying most tickets paying nothing makes a positive average payout impossible

[STAR] THE PROBLEM — annotated

Suppose a club sells 200 raffle tickets for \$2 each. One ticket wins \$100, four tickets win \$25 each, 20 tickets win \$5 each, and the other tickets win nothing. Let X be the buyer's net profit from one randomly chosen ticket: prize minus ticket cost.

Priya: *The average gross payout is \$1.50 per ticket, so buying one is a good deal.*

Mateo: *Most tickets win nothing.*

Ticket net-profit distribution

Prize	Net x	$P(X = x)$
Top prize	98	1/200
Second prize	23	4/200
Small prize	3	20/200
No prize	−2	175/200

FIRST TAKE — one sentence before you work the parts

Would you buy one ticket if your goal were to make money? Use one number or outcome from the problem.

Key: Not graded. Students may focus on the top prize or on the 175 losing tickets; the distribution should connect those views.

[BOOK] DISTRIBUTIONS AND ONE OUTCOME (keep on the page)

A discrete random variable assigns a number to each random outcome. Its distribution lists every possible value with a probability between 0 and 1; all probabilities add to 1. Add the probabilities of the outcomes in an event. Net profit is prize minus ticket cost. An average can differ from the most common outcome; large prizes can raise it, and it does not promise one buyer's result.

DOK 1 (a) State the four possible net profits X . Which is most likely?

Key: The possible net profits are 98, 23, 3, and −2 dollars. The most likely value is $X = -2$, because 175 of the 200 tickets pay no prize.

DOK 2 (b) Check that the listed probabilities add to 1. Find the chance that a ticket loses money.

Key: $(1 + 4 + 20 + 175)/200 = 1$, with every probability between 0 and 1. Only the no-prize ticket loses money, so $P(X < 0) = 175/200 = 0.875 = 87.5\%$.

DOK 3 (c) Can Priya's average payout and Mateo's statement both be true? Use the distribution to explain. Decide whether Priya's good-deal claim follows for a buyer trying to make money, using the ticket cost and possible net profits. Write 3–4 sentences.

Key: Both statements can be true: a few large prizes raise the average payout even though 175 of 200 tickets pay nothing. An average does not promise that one buyer receives \$1.50; the only net profits are \$98, \$23, \$3, and −\$2. Priya's claim does not follow for making money because the \$1.50 average payout is below the \$2 cost and the chance of losing \$2 is 87.5%. A buyer could still choose to support the club, but that is a different reason from expecting to make money.